

Technical Documentations

1. Introduction

Retrieval-Augmented Generation (RAG) is a framework that optimizes the output of a Large Language Model (LLM) by referencing an authoritative, external knowledge base before generating a response. This project is needed to address the inherent limitations of standard LLMs, such as hallucinations and outdated training data. By grounding the model in a specific Customer Support Knowledge Base, the system provides accurate, fact-based answers regarding company policies.

2. System Architecture Explanation

The system follows a modular architecture consisting of a Data Ingestion Pipeline, a Vector Database, and a Stateful Orchestration Layer:

- **Data Ingestion:** Documents are parsed using `UnstructuredWordDocumentLoader` and broken into manageable segments.
- **Vector Database:** High-dimensional embeddings are stored in ChromaDB, allowing for efficient semantic retrieval.
- **Orchestration:** LangGraph manages the flow of information between the retriever, the LLM (flan-t5-base), and the human interface.

3. Design Decisions

- **Chunk Size Choice:** A `chunk_size` of 500 characters with a 50-character overlap was selected to ensure each segment contains sufficient context while remaining small enough for precise retrieval.
- **Embedding Strategy:** The `all-MiniLM-L6-v2` model was chosen for its balance of speed and performance in mapping semantic similarity.
- **Retrieval Approach:** A `k=3` search parameter is used to retrieve the three most relevant context chunks to balance information density with the model's token limits.

- Prompt Design Logic: The prompt uses strict grounding, instructing the model to "Answer ONLY using the context below," which minimizes creative drift.

4. Workflow Explanation

The system utilizes LangGraph to define a directed acyclic graph (DAG) where each step is a discrete node:

- process Node: Responsible for retrieving documents and generating an initial LLM response.
- hitl (Human-in-the-Loop) Node: Pauses the graph to allow a human operator to provide input if the AI is uncertain.
- output Node: Formats and delivers the final response to the user.

5. Conditional Logic

The system implements Confidence-Based Routing to determine the path of a query:

- Logic Gate: After the process node, the `route_decision` function checks the confidence score.
- Automated Path: If confidence is ≥ 0.6 (meaning relevant documents were found), the graph routes to the output node.
- Escalation Path: If confidence is < 0.6 , the system identifies a failure in retrieval and routes to the hitl node.

6. HITL Implementation

Human-in-the-Loop (HITL) serves as a fail-safe for the AI:

- Role: When the retriever returns no relevant information, the system triggers an escalation, prompting a human to "Enter human response".

- Benefits: Ensures 100% accuracy for customer-facing answers and prevents the AI from providing incorrect "hallucinated" info.
- Limitations: Manual intervention increases response latency compared to a fully automated system.

7. Challenges & Trade-offs

- Retrieval Accuracy vs. Speed: Increasing the number of retrieved chunks (\$k\$) improves context but slows down the inference time of the local_llm.
- Chunk Size vs. Context Quality: Larger chunks provide more context but may include irrelevant "noise" that confuses the flan-t5-base model.

8. Testing Strategy

The system was tested using real-world policy queries:

- Direct Success: Queries like "Refund Policy" successfully retrieved the correct context and provided a grounded answer.
- Escalation Success: Ambiguous or "out-of-context" queries successfully triggered the Escalation Triggered warning and requested human input.

9. Future Enhancements

- Memory Integration: Adding WindowMemory to allow the graph to reference previous turns in the conversation.
- Feedback Loop: Allowing the human responses to be automatically added back into the ChromaDB vector store to "teach" the AI for future queries.
- Multi-Document Support: Expanding the loader to handle entire directories of mixed PDF and Word files.